

FACTSHEET FS07 · Size

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

Do smaller companies beat bigger ones?

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Size premium (Fama-French SMB)	Banz (1981)	Size
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (smallest names in the universe)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

The claim. Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -4.3% to +4.3% a year and was positive in 1 of the six; no universe passes the 2.87 gate. The long-only book (smallest names in the universe) had a higher Sharpe than the equal-weight universe in 1 of the six (UK), with alphas of -6.2% to +6.4%, passing the gate in UK. **What goes wrong (§10):** the main drags are a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Size; its returns sit closest to the low risk cluster of the signal library. **360° view (§12):** nearest neighbours Momentum 12-1, Near 52-week high; after those neighbours and the market, nothing is left (alpha |t| < 2 everywhere): the signal is redundant.

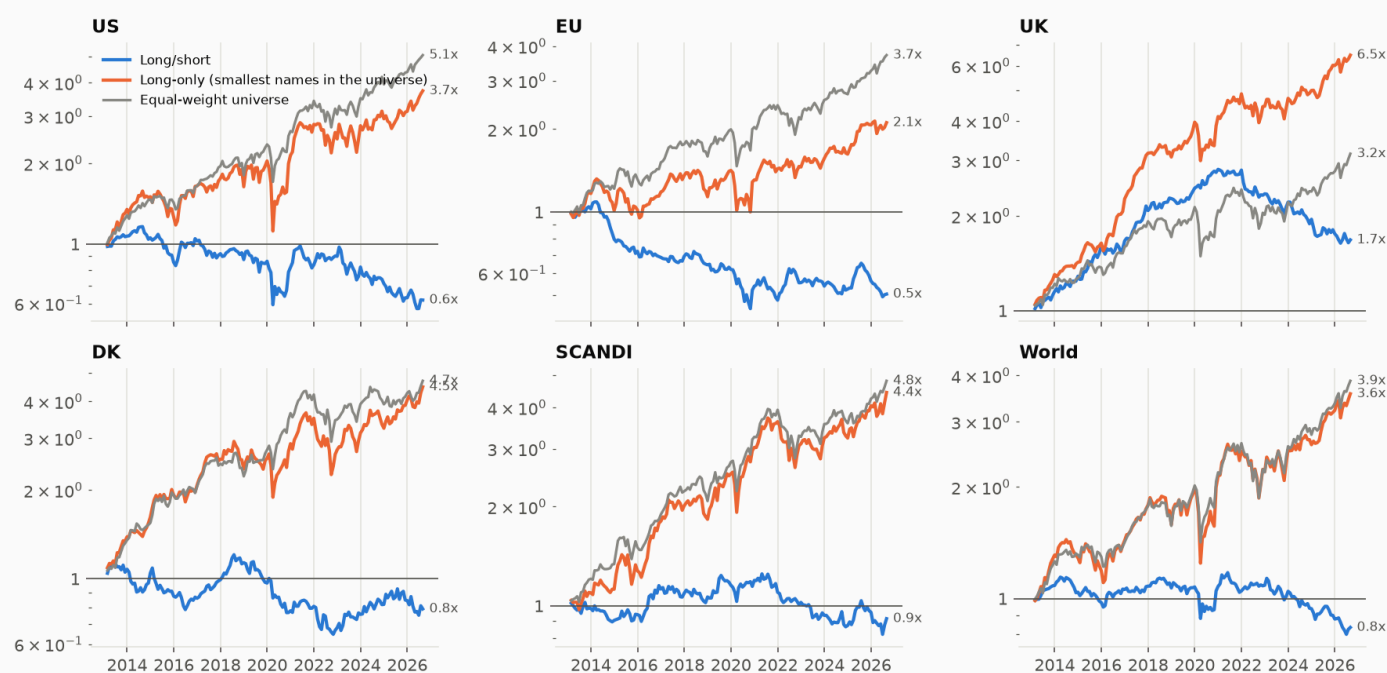
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long- only CAGR	Long- only Sharpe	Long- only max DD	EW CAGR	EW Sharpe	Size- weighted CAGR
US	499 / 10	-2.2%	-0.55	-10.4% (-2.8)	0.62	+10.2%	0.53	-45%	+12.7%	0.85	+14.9%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
EU	260 / 10	-4.3%	-1.19	-6.1% (-1.5)	0.16	+5.7%	0.41	-29%	+10.1%	0.74	+10.2%
UK	330 / 10	+4.3%	1.52	+3.1% (1.0)	0.13	+14.8%	1.06	-25%	+8.8%	0.67	+7.8%
DK	20 / 3	-0.8%	-0.23	-1.8% (-0.5)	0.08	+11.7%	0.71	-38%	+12.1%	0.83	+10.2%
SCANDI	70 / 5	-0.1%	-0.02	-2.5% (-0.9)	0.19	+11.6%	0.71	-29%	+12.3%	0.89	+11.0%
World	1088 / 10	-0.9%	-0.38	-4.5% (-2.0)	0.32	+9.8%	0.58	-37%	+10.5%	0.71	+10.3%

Monthly, local currency (World in USD). L/S = equal-weighted smallest names in the universe minus largest names, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Size on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjerner · Project 10 factsheets

2. Strategy description

Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018).

Why it might work. Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January.

Rule	Original (Banz (1981))	This factsheet
Signal	Market capitalisation	US: market cap (Sharadar). Elsewhere: 63-day average traded value (price × volume), a proxy
Portfolio	Size deciles or SMB (2×3)	Extreme quantile L/S within the universe, monthly
Weighting	Value weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as run_wiki_pit.py / run_eu_pit.py), with a short list of verified ticker renames (e.g. NZYM-B → NSIS-B, WDH → DEMANT, DAI → MBG). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
- **Data gap:** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.

4. Signal creation

$SIZE_{i,t}$ = log market cap (US) or log 63-day average traded value (elsewhere) at month-end t ; World ranks within each region first. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- Long/short** = smallest names in the universe minus largest names; **long-only** = smallest names in the universe; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (S&P 500 members, point-in-time), median 499 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-2.2%	-3.5%	16.4%	-0.13 [-0.58, 0.33]	-0.30	-51%	47%	-0.55	0.62	-10.4% (-2.8)	9.5%
Long-only (net)	+12.6%	+10.2%	23.9%	0.53 [0.10, 1.05]	0.67	-45%	61%	2.41	1.42	-6.2% (-2.7)	13.6%
Smallest names in the universe (gross)	+12.8%	+10.4%	23.9%	0.54 [0.11, 1.06]	0.68	-45%	61%	2.45	1.42	-6.0% (-2.6)	13.6%
Largest names (gross)	+14.5%	+14.4%	13.7%	1.05 [0.60, 1.60]	1.75	-26%	68%	5.33	0.80	+3.9% (2.2)	7.7%
EW universe	+13.2%	+12.7%	15.5%	0.85 [0.42, 1.42]	1.30	-27%	68%	4.26	1.00	–	9.2%
Size-weighted universe	+14.9%	+14.9%	14.2%	1.05 [0.60, 1.61]	1.73	-24%	69%	5.29	0.86	+3.6% (2.3)	8.3%

EU (11 national blue-chip indices, point-in-time), median 260 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-4.3%	-4.9%	11.8%	-0.37 [-1.00, 0.17]	-0.60	-59%	39%	-1.19	0.16	-6.1% (-1.5)	6.2%
Long-only (net)	+6.9%	+5.7%	16.9%	0.41 [-0.04, 0.92]	0.52	-29%	57%	1.66	1.03	-4.1% (-1.6)	10.2%
Smallest names in the universe (gross)	+7.1%	+5.8%	16.9%	0.42 [-0.03, 0.93]	0.54	-29%	57%	1.70	1.03	-4.0% (-1.6)	10.2%
Largest names (gross)	+10.9%	+10.4%	13.9%	0.78 [0.30, 1.34]	1.18	-28%	61%	3.22	0.87	+1.6% (0.8)	8.3%
EW universe	+10.7%	+10.1%	14.5%	0.74 [0.28, 1.29]	1.13	-26%	62%	3.18	1.00	–	8.5%
Size-weighted universe	+10.6%	+10.2%	13.7%	0.77 [0.32, 1.29]	1.21	-25%	58%	3.37	0.91	+0.9% (0.7)	7.9%

UK (FTSE 350, point-in-time), median 330 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.3%	+3.9%	9.9%	0.43 [-0.18, 1.07]	0.62	-42%	55%	1.52	0.13	+3.1% (1.0)	5.0%
Long-only (net)	+14.8%	+14.8%	13.9%	1.06 [0.55, 1.74]	1.78	-25%	67%	4.29	0.89	+6.4% (4.1)	8.2%
Smallest names in the universe (gross)	+15.2%	+15.2%	13.9%	1.09 [0.57, 1.77]	1.83	-24%	68%	4.39	0.89	+6.7% (4.3)	8.2%
Largest names (gross)	+10.1%	+9.7%	12.9%	0.79 [0.24, 1.44]	1.18	-26%	64%	2.97	0.76	+2.9% (1.2)	7.6%
EW universe	+9.5%	+8.8%	14.2%	0.67 [0.19, 1.27]	0.94	-30%	59%	2.84	1.00	–	8.8%
Size-weighted universe	+8.4%	+7.8%	12.9%	0.65 [0.16, 1.25]	0.90	-28%	62%	2.68	0.83	+0.4% (0.2)	7.8%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-0.8%	-1.7%	13.7%	-0.06 [-0.53, 0.46]	-0.18	-46%	53%	-0.23	0.08	-1.8% (-0.5)	8.2%
Long-only (net)	+12.7%	+11.7%	17.8%	0.71 [0.18, 1.36]	0.99	-38%	58%	2.77	1.05	-0.6% (-0.2)	10.6%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Smallest names in the universe (gross)	+12.9%	+11.9%	17.8%	0.73 [0.20, 1.38]	1.02	-38%	58%	2.82	1.05	-0.3% (-0.1)	10.6%
Largest names (gross)	+13.1%	+12.3%	17.0%	0.77 [0.26, 1.33]	1.24	-29%	60%	3.05	0.97	+0.9% (0.4)	8.9%
EW universe	+12.6%	+12.1%	15.2%	0.83 [0.31, 1.44]	1.28	-33%	66%	3.31	1.00	–	8.8%
Size-weighted universe	+11.1%	+10.2%	16.4%	0.68 [0.11, 1.33]	0.94	-40%	62%	2.55	0.93	-0.6% (-0.2)	10.7%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-0.1%	-0.7%	11.0%	-0.01 [-0.47, 0.45]	-0.09	-34%	47%	-0.02	0.19	-2.5% (-0.9)	6.3%
Long-only (net)	+12.6%	+11.6%	17.6%	0.71 [0.25, 1.24]	1.05	-29%	58%	2.99	1.12	-1.6% (-0.9)	10.5%
Smallest names in the universe (gross)	+12.7%	+11.7%	17.6%	0.72 [0.26, 1.24]	1.06	-29%	58%	3.01	1.12	-1.5% (-0.9)	10.4%
Largest names (gross)	+12.3%	+11.8%	14.2%	0.86 [0.41, 1.40]	1.33	-25%	65%	3.98	0.93	+0.5% (0.4)	8.8%
EW universe	+12.6%	+12.3%	14.2%	0.89 [0.37, 1.52]	1.38	-29%	66%	3.64	1.00	–	8.4%
Size-weighted universe	+11.4%	+11.0%	13.5%	0.84 [0.37, 1.41]	1.32	-26%	63%	3.59	0.90	-0.0% (-0.0)	8.1%

World (US + UK + EU, point-in-time), median 1088 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-0.9%	-1.3%	9.3%	-0.10 [-0.60, 0.39]	-0.20	-32%	50%	-0.38	0.32	-4.5% (-2.0)	5.3%
Long-only (net)	+11.3%	+9.8%	19.6%	0.58 [0.12, 1.13]	0.79	-37%	59%	2.46	1.17	-1.9% (-1.3)	11.6%
Smallest names in the universe (gross)	+11.6%	+10.1%	19.6%	0.59 [0.14, 1.14]	0.81	-37%	59%	2.51	1.17	-1.7% (-1.1)	11.6%
Largest names (gross)	+11.9%	+11.4%	14.1%	0.84 [0.37, 1.43]	1.30	-28%	66%	3.65	0.85	+2.3% (2.0)	8.4%
EW universe	+11.3%	+10.5%	16.0%	0.71 [0.25, 1.27]	1.02	-30%	62%	3.07	1.00	–	9.7%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+11.0%	+10.3%	14.8%	0.74 [0.28, 1.31]	1.10	-28%	61%	3.19	0.91	+0.7% (0.8)	8.9%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+7.0%	+2.7%	+15.2%	+7.8%	-1.8%	+8.5%
2014	-1.4%	-22.3%	+11.8%	-2.9%	-8.0%	-4.0%
2015	-19.0%	-9.6%	+24.4%	-16.6%	+4.1%	-6.2%
2016	+20.1%	-5.0%	+4.1%	-1.7%	+19.6%	+8.4%
2017	-10.3%	+1.0%	+24.8%	+15.6%	-2.9%	+2.1%
2018	-4.9%	-7.0%	+9.1%	+14.3%	-2.6%	-1.2%
2019	-1.4%	-3.8%	+8.0%	-14.7%	+8.5%	+0.6%
2020	-5.1%	-13.1%	+14.7%	-19.5%	+3.5%	+0.3%
2021	+2.1%	-11.0%	-0.6%	+3.6%	-8.0%	-2.8%
2022	+7.6%	+15.8%	-15.9%	-15.4%	-6.8%	+3.6%
2023	-8.8%	-2.8%	-5.9%	+11.0%	-8.7%	-3.7%
2024	-13.3%	-2.5%	-11.8%	+8.1%	-6.8%	-10.2%
2025	-11.5%	+7.0%	-10.1%	+3.0%	+9.6%	-6.3%
2026	-1.9%	-10.4%	-4.3%	-6.3%	-4.4%	-5.1%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+34.6%	+16.8%	+30.4%	+38.8%	+9.5%	+31.1%
2014	+11.5%	-13.2%	+9.4%	+16.8%	+2.6%	-1.8%
2015	-16.0%	+0.4%	+15.0%	+18.7%	+13.8%	-7.7%
2016	+30.7%	+16.3%	+38.4%	+8.1%	+41.4%	+20.8%
2017	+6.7%	+14.9%	+39.6%	+22.5%	+11.5%	+24.5%
2018	-7.0%	-16.3%	-2.8%	-7.4%	-9.1%	-11.5%
2019	+24.8%	+24.7%	+29.0%	+8.8%	+37.5%	+26.0%
2020	+5.0%	-7.6%	+4.2%	+12.1%	+20.2%	+7.4%
2021	+30.4%	+9.8%	+18.0%	+22.0%	+19.7%	+16.7%
2022	-11.7%	+1.4%	-8.7%	-21.9%	-14.4%	-11.9%
2023	+12.0%	+7.5%	+5.9%	+14.1%	+4.4%	+14.7%
2024	+6.1%	+3.7%	+6.0%	+3.7%	-0.2%	+3.1%

Year	US	EU	UK	DK	SCANDI	World
2025	+6.6%	+27.6%	+20.0%	+24.8%	+25.3%	+23.8%
2026	+19.2%	+1.9%	+7.9%	+10.8%	+9.7%	+10.8%

7. Trading record

The rebalance log per universe is in `results/xs_FS07_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	50 / 49	9%	4%	47%	-24.2%	2020-03
EU	163	26 / 26	6%	6%	39%	-9.4%	2020-07
UK	163	33 / 33	15%	5%	55%	-8.7%	2022-01
DK	163	7 / 6	10%	6%	53%	-12.6%	2020-03
SCANDI	163	14 / 14	4%	6%	47%	-8.6%	2024-10
World	163	109 / 108	10%	5%	50%	-14.2%	2020-03

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: smallest names in the universe	Short: largest names
US	TTD, TAP, BLDR, NCLH, MOS, CAG, AOS, ARE	NVDA, AAPL, GOOGL, MSFT, AMZN, AVGO, META, TSLA
EU	NYR.BR, OBEL.BR, AGFB.BR, GTC.WA, KER.WA, TESB.BR, BPOST.BR, ONTEX.BR	NOVO-B.CO, ASML.AS, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST
UK	RTW.L, BPCR.L, PEY.L, MNTN.L, MTLN.L, HANA.L, PRN.L, NAS.L	AZN.L, HSBA.L, RR.L, SHEL.L, GLEN.L, AAL.L, RIO.L, BP.L
DK	NDA-DK.CO, BAVA.CO, GN.CO, RBREW.CO, AMBU-B.CO, ISS.CO, MAERSK-A.CO, JYSK.CO	NOVO-B.CO, DSV.CO, VWS.CO, MAERSK-B.CO, DANSKE.CO, NSIS-B.CO, GMAB.CO, PNDORA.CO
SCANDI	KALMAR.HE, QTCOM.HE, KEMIRA.HE, HIAB.HE, TYRES.HE, TIETO.HE, MANTA.HE, HUH1V.HE	NOVO-B.CO, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST, DSV.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-2.6%	-1.12	0.78	mkt +0.22 (+3.8), size +0.44 (+3.6), value +0.75 (+6.7), momentum -0.22 (-2.5), low_risk -0.37 (-2.3), quality -0.62 (-4.1)
EU	French Europe 5F + WML	-5.2%	-1.60	0.20	SMB +0.47 (+3.4), HML +0.36 (+2.1)
UK	JKP GBR 7 themes	+5.4%	2.29	0.41	size +0.97 (+8.2), value -0.46 (-3.5)
DK	JKP DNK 7 themes	-1.9%	-0.52	0.15	size +0.53 (+3.0), low_risk +0.32 (+2.3)
SCANDI	French Europe 5F + WML	-0.7%	-0.30	0.10	SMB +0.47 (+2.4)
World	JKP World 7 themes	-5.5%	-2.71	0.60	mkt +0.12 (+3.4), size +0.63 (+5.2), value +0.42 (+4.0), short_term_reversal +0.35 (+2.4)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+2.1%	1.16	0.94	mkt +1.07 (+23.7), value +0.69 (+7.1), momentum -0.25 (-4.9), low_risk -0.33 (-2.3), quality -0.40 (-2.7), short_term_reversal +0.41 (+2.2)
EU	French Europe 5F + WML	+0.9%	0.37	0.64	Mkt-RF +0.74 (+10.4), HML +0.43 (+2.5)
UK	JKP GBR 7 themes	+11.1%	4.38	0.70	mkt +0.51 (+8.8), size +0.31 (+2.9), low_risk -0.36 (-2.6)
DK	JKP DNK 7 themes	+4.1%	1.05	0.61	mkt +0.77 (+9.8), momentum -0.20 (-2.2)
SCANDI	French Europe 5F + WML	+7.1%	2.30	0.51	Mkt-RF +0.76 (+8.9)
World	JKP World 7 themes	+2.8%	1.15	0.87	mkt +0.77 (+14.3), size +0.33 (+3.3), value +0.51 (+4.4), low_risk -0.23 (-2.2)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books $\times$ 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -10.4% to +3.1%, $|t|$ past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 1 of the six; alpha -6.2% to +6.4%, passing in 1 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.31, EU 0.10, UK 0.94, DK 0.42, SCANDI 0.49, World 0.36.
- **Publication decay:** gross L/S -5.9% to +14.1% a year in 2013–19 against -4.4% to -1.2% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	-0.9%	0.53	+6.9%	-7.8%	-0.3%	-0.9%	+0.1%	14%	2019-08 -10%, 2015-07 -6%, 2015-12 -6%
EU	-5.9%	0.07	+0.7%	-6.6%	-0.3%	-4.5%	-1.4%	13%	2014-05 -6%, 2014-10 -6%, 2015-11 -5%
UK	+14.1%	-0.19	-2.2%	+16.4%	-0.4%	+9.4%	+4.7%	19%	2018-04 -4%, 2016-10 -4%, 2017-12 -3%
DK	+0.8%	-0.08	-1.2%	+2.0%	-0.3%	-0.6%	+1.4%	15%	2015-02 -8%, 2014-02 -7%, 2014-05 -7%
SCANDI	+3.1%	0.17	+2.7%	+0.4%	-0.2%	-0.6%	+3.6%	9%	2015-09 -5%, 2013-10 -4%, 2017-07 -4%
World	+1.8%	0.23	+2.5%	-0.6%	-0.3%	+0.4%	+1.5%	14%	2019-08 -5%, 2014-11 -3%, 2018-12 -3%

On average across the six universes the gross spread was +2.2% a year, of which the market exposure (beta +0.12) contributed +1.6%; the beta-adjusted spread (CAPM alpha) was +0.6%. Costs took 0.3% a year at 14% monthly turnover across both legs. The long leg beat the universe by +0.5% and the short leg lagged it by +1.6% a year, so most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +0.3%, +0.2%, -0.9% a year.

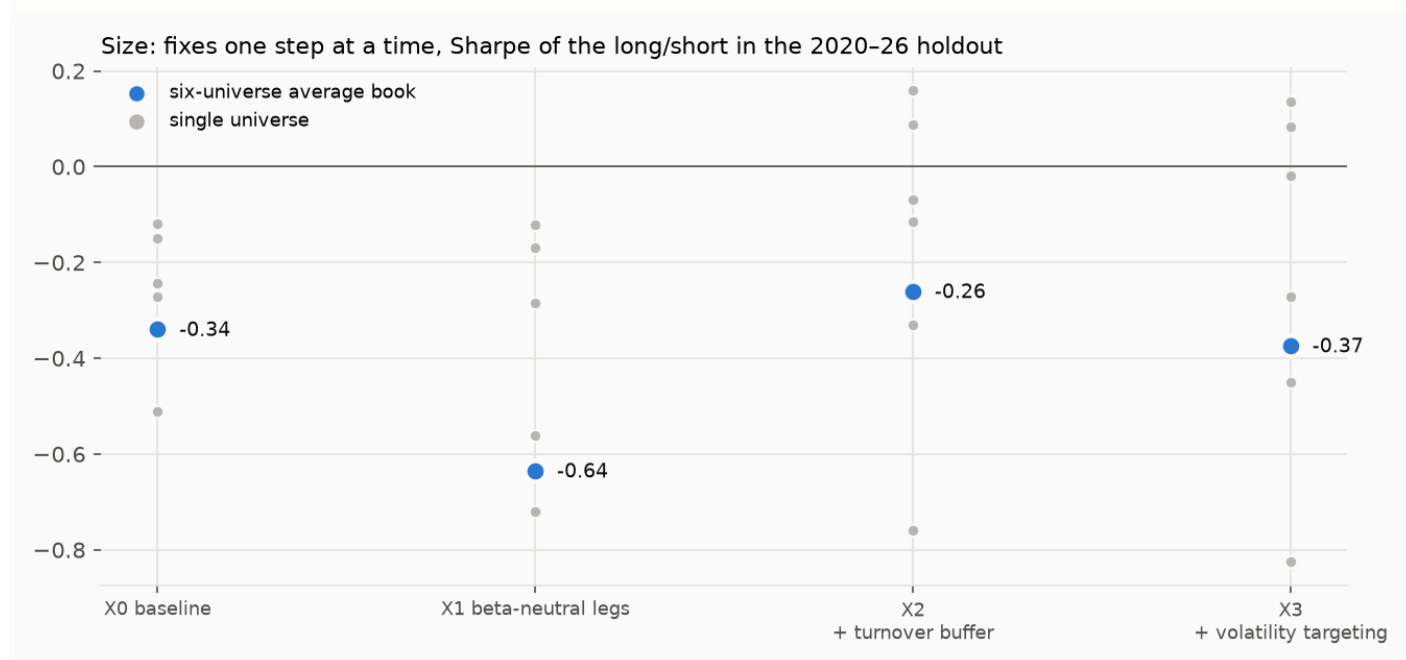
10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.19	–	-0.34	–	–	-3.0%	-27%	-6.1% (-2.0)	0.3%

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X1 beta-neutral legs	0.25	+0.26 (p 0.035)	-0.64	-0.30 (p 0.994)	1 of 6	-5.3%	-34%	-7.3% (-2.3)	0.4%
X2 + turnover buffer	0.37	+0.23 (p 0.081)	-0.26	+0.37 (p 0.020)	5 of 6	-1.6%	-18%	-2.7% (-1.2)	0.1%
X3 + volatility targeting	0.34	-0.25 (p 0.753)	-0.37	-0.11 (p 0.882)	1 of 6	-3.0%	-22%	-4.7% (-1.5)	0.1%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



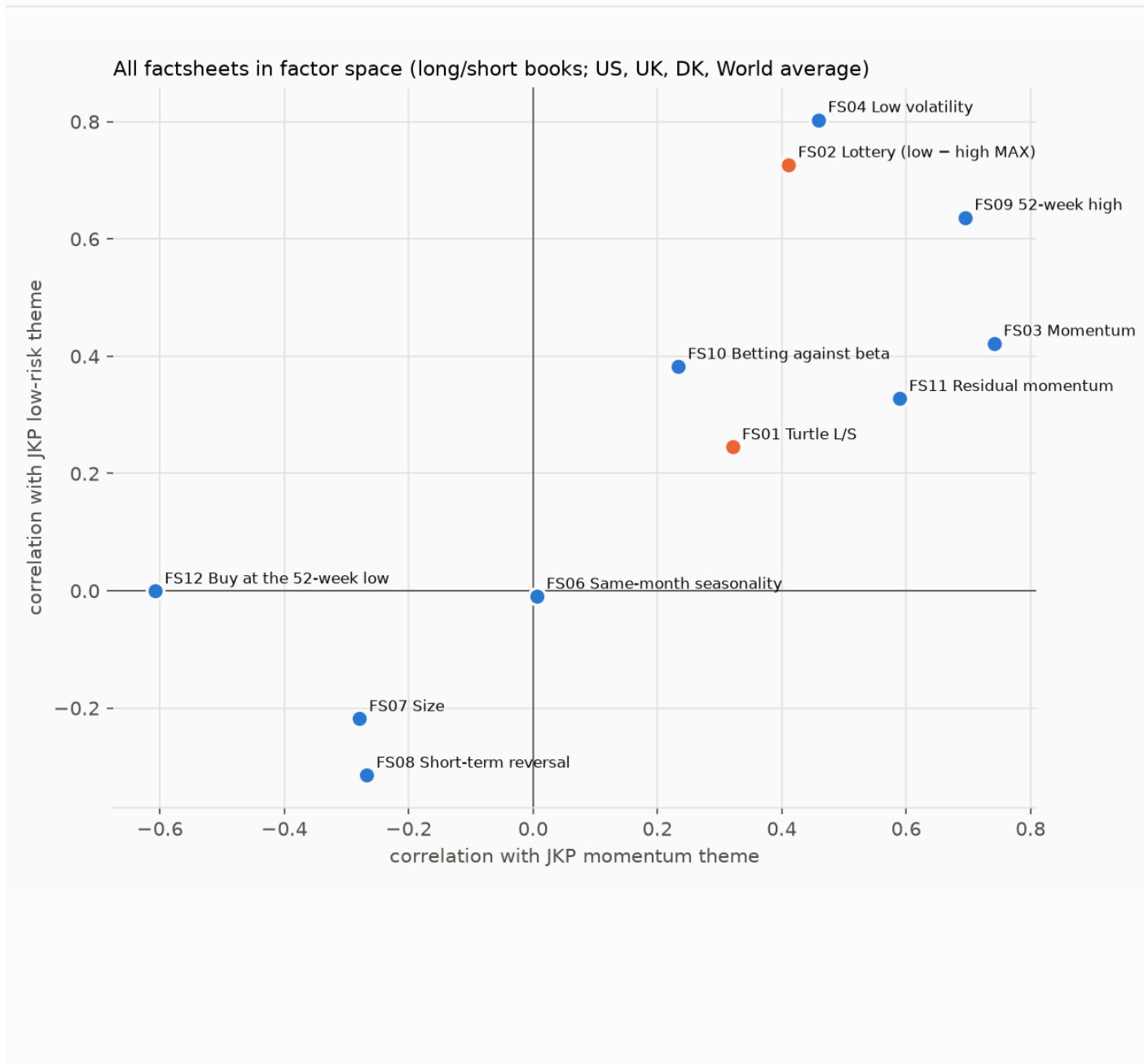
- **X1 beta-neutral legs:** +0.26 design, -0.30 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.23 design, +0.37 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.25 design, -0.11 holdout; hurts in both windows.

Where it ends: holdout Sharpe -0.34 → -0.37, return -3.0% → -3.0% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Size
Series family	Size
Academic style	Size premium (Fama-French SMB)
Signal	Price only (daily total returns)

Dimension	Size
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 14% of the two legs replaced monthly
Market exposure	L/S beta 0.08 to 0.62
Payoff shape	Treynor–Mazuy γ -2.21 to 1.24 (t -2.6 to 0.8); worst 10% of market months -3.2% to -0.7% a month, best 10% -0.9% to +5.2%
Nearest factor theme	size (3 of 6 universes), quality (1 of 6 universes)
Nearest library signals (returns)	Momentum 12-1 (3 universes), Volatility (3 universes), Near 52-week high (2 universes)
Economic rationale	Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January.
Publication	Banz 1981; Fama & French 1992–93; 'Is size dead?' (van Dijk 2011); 'Size matters, if you control your junk' (Asness et al. 2018)
Where it fits	In the low risk cluster of the library; fully explained by its neighbours.

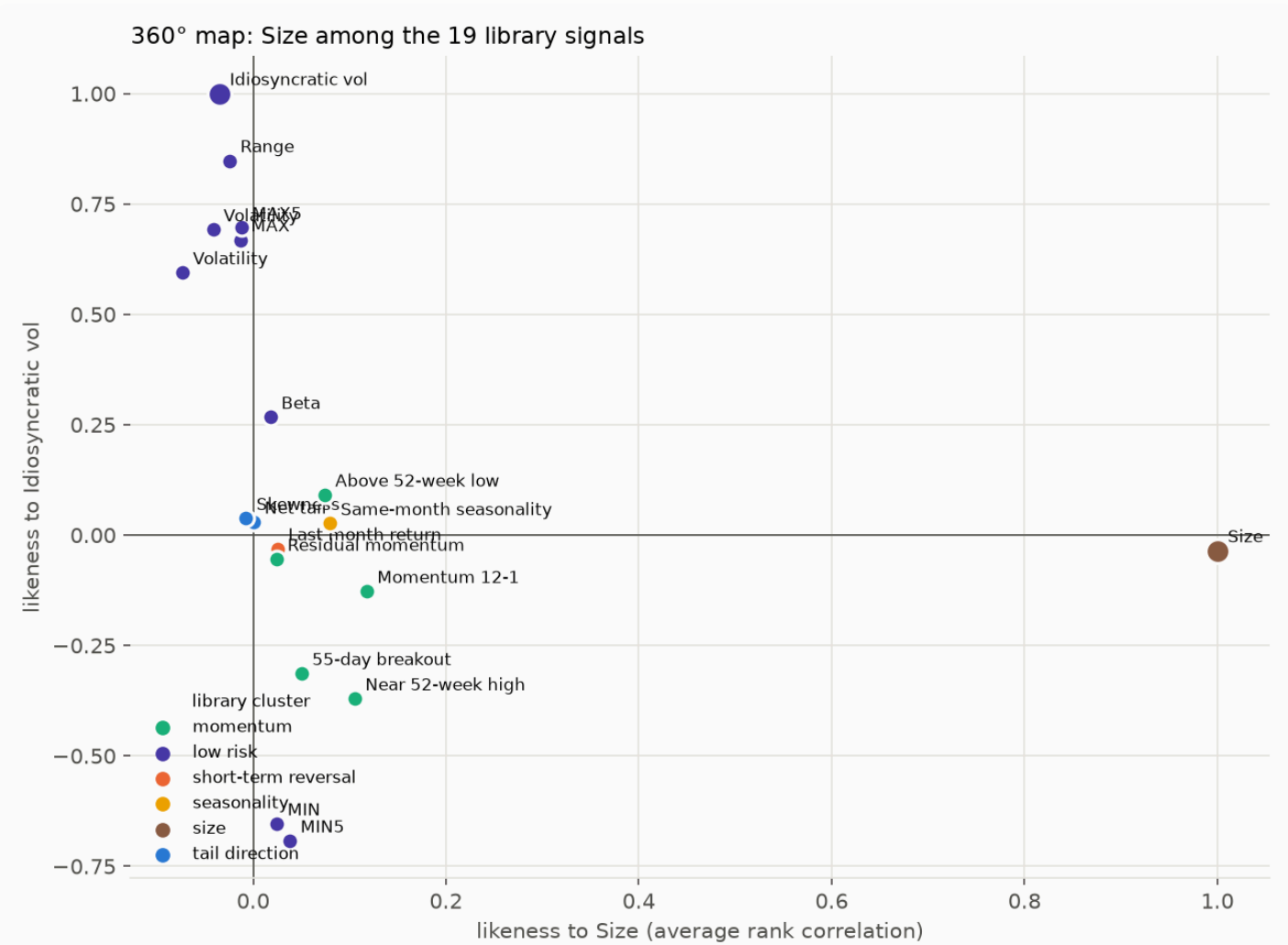


12. 360° view

The signal among the 19 price signals of the shared library (code/library.py), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Idiosyncratic vol**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ -2
Size (this strategy)	size	+1.00	+1.00	+0.1%	-3.8% to +9.9%	1 / 0
Near 52-week high	momentum	+0.11	+0.36	+3.7%	+9.3% to +22.3%	6 / 0
Momentum 12-1	momentum	+0.12	+0.33	+7.2%	+4.3% to +19.4%	5 / 0
Volatility (252 days)	low risk	-0.07	-0.33	+2.4%	-12.9% to -4.8%	0 / 3
55-day breakout	momentum	+0.05	+0.32	+1.4%	+4.1% to +13.0%	4 / 0
Volatility	low risk	-0.04	-0.32	+1.1%	-13.7% to -7.9%	0 / 5
Range (MAX-MIN)	low risk	-0.02	-0.31	-0.7%	-15.4% to -4.8%	0 / 4
MIN5	low risk	+0.04	+0.30	-0.9%	+2.3% to +13.7%	4 / 0
MIN (worst day)	low risk	+0.02	+0.30	-0.7%	+3.3% to +11.1%	4 / 0
Idiosyncratic vol (mirror)	low risk	-0.04	-0.29	-3.6%	-15.5% to -4.8%	0 / 4
Beta	low risk	+0.02	-0.28	+4.4%	-12.6% to -4.2%	0 / 4
MAX (best day)	low risk	-0.01	-0.25	+0.1%	-13.3% to -2.0%	0 / 2
MAX5	low risk	-0.01	-0.24	+0.6%	-11.3% to -3.5%	0 / 2
Last month return	short-term reversal	+0.03	+0.22	+1.8%	-3.2% to +11.0%	2 / 0
Above 52-week low	momentum	+0.07	+0.15	+7.5%	+5.2% to +13.6%	4 / 0
Residual momentum	momentum	+0.02	+0.11	+6.4%	+8.0% to +12.2%	6 / 0
Same-month seasonality	seasonality	+0.08	+0.08	-1.9%	-9.4% to +6.6%	0 / 1
Net tail (MAX+MIN)	tail direction	+0.00	+0.07	-0.9%	-5.2% to +5.2%	0 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ -2
Skewness	tail direction	-0.01	-0.06	-0.4%	-6.7% to +3.9%	0 / 2



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with t ≥ 2
US	+3.3%	1.27	0.72	Momentum 12-1 +0.39 (+4.8)
EU	+1.4%	0.40	0.17	Idiosyncratic vol -0.30 (-2.2)
UK	-1.5%	-0.57	0.14	Same-month seasonality -0.15 (-3.8), Momentum 12-1 -0.12 (-2.2)
DK	+0.6%	0.20	0.21	Above 52-week low +0.22 (+2.2), Momentum 12-1 +0.24 (+2.3)
SCANDI	+0.7%	0.21	0.14	Volatility -0.27 (-2.0)
World	+1.6%	0.80	0.54	MIN5 -0.19 (-2.4)

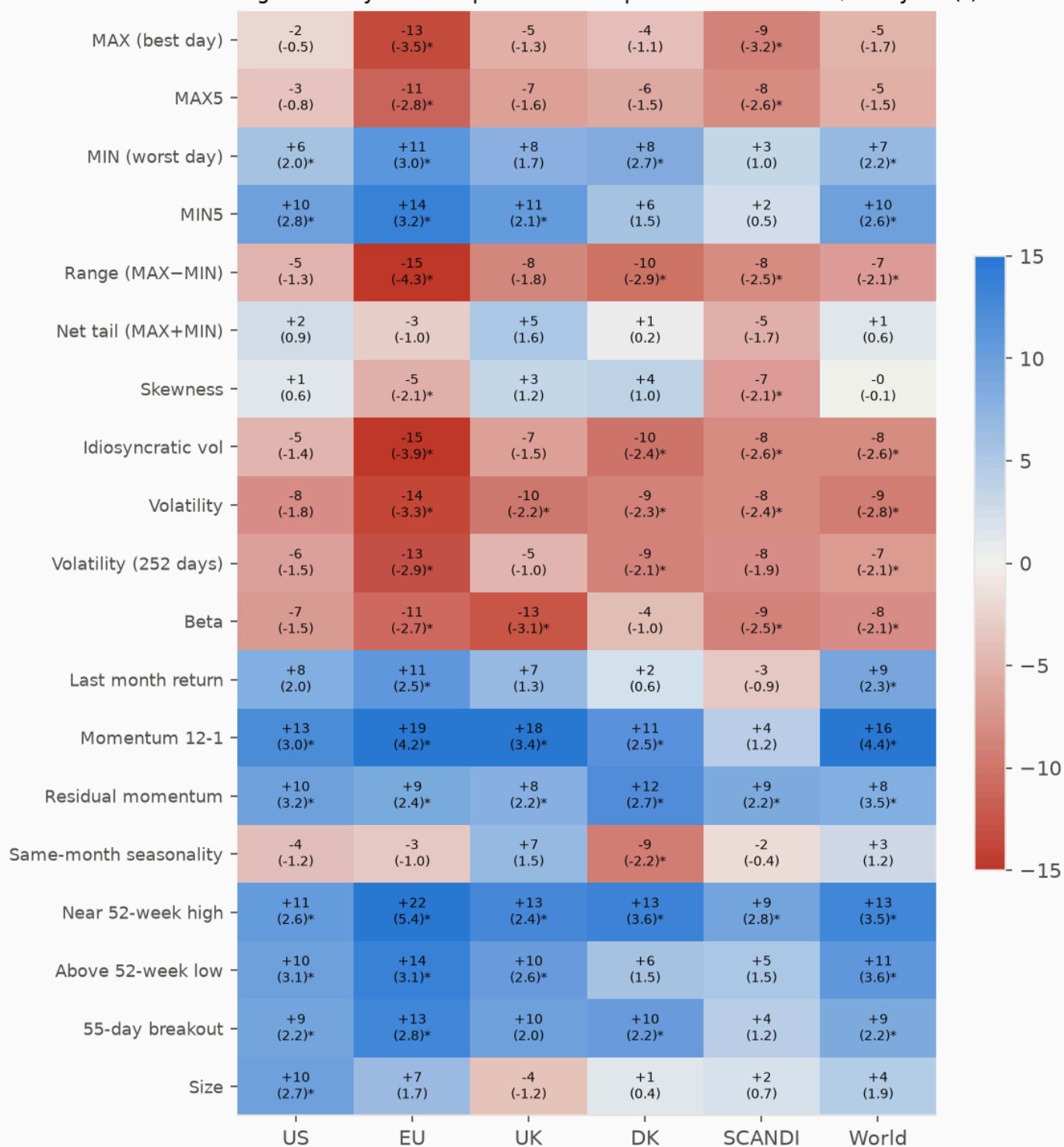
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Size	12.3%	13.0%	11.6%
middle	11.1%	12.5%	10.8%
high Size	11.8%	11.8%	9.6%
	low Idiosyncratic vol	middle	high Idiosyncratic vol

Holding Idiosyncratic vol fixed (down a column), moving from low to high Size changes the return by -0.5, -1.2 and -1.9 points a year; holding Size fixed (along a row), moving from low to high Idiosyncratic vol changes it by -0.7, -0.3 and -2.2. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -6% vs +11% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +0.1%, +0.3%, +0.2%, -0.9% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
5. **Data gap.** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.
6. **Not a registered trial.** Descriptive; the fix ladder is counted inside FSO3–FS12 (25 trials).

14. Academic references

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15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FSO7 → build_xs.py FSO7.

Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.